

Sandeep Kumar Karnati

Ohio, USA

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PROFESSIONAL SUMMARY

Full Stack Developer with 4+ years of expertise in Node.js, Express.js, RESTful APIs, and microservices architecture. Proven track record in optimizing database performance and implementing secure authentication mechanisms, reducing API response times by 50% and enhancing security. Experienced in deploying applications on AWS and managing containerized environments with Docker and Kubernetes. Eager to leverage backend development skills to build scalable and efficient systems.

TECHNICAL SKILLS

- **Backend:** Node.js, Express.js, NestJS, RESTful APIs, GraphQL, WebSockets
- **Frontend:** Angular, React.js, Next.js, Redux, Tailwind CSS, Bootstrap
- **Databases:** MongoDB, PostgreSQL, MySQL, Redis
- **DevOps & Cloud:** AWS (Lambda, EC2, RDS, S3, CloudWatch), Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitHub Actions)
- **Authentication & Security:** JWT, OAuth2, RBAC, API Gateway, AWS IAM
- **Messaging & Queues:** Kafka, RabbitMQ, BullMQ, Redis Pub/Sub
- **Testing & Performance:** Jest, Mocha, Cypress, Postman, Apache JMeter

WORK EXPERIENCE

Procter & Gamble (P&G)

Jun 2024 - Present

Full Stack Developer

Cincinnati, Ohio, USA

- Developed and deployed RESTful APIs using Node.js and Express.js, enabling seamless data exchange between internal applications and third-party services. This improved system integration and reduced API latency by 40%.
- Implemented a microservices architecture using NestJS, breaking down a monolithic application into independent services. This enhanced system scalability and enabled better maintenance and faster deployments.
- Created a real-time analytics dashboard using Angular and WebSockets, allowing internal teams to track production and logistics data instantly. This reduced report generation time from 2 hours to real-time insights.
- Optimized database queries and indexing for PostgreSQL, reducing API response time by 50% by restructuring SQL queries and adding necessary indexes.
- Integrated Kafka for asynchronous messaging, ensuring that order and inventory services communicated seamlessly. This reduced order processing delays by 30%.
- Implemented secure authentication and authorization mechanisms using JWT and OAuth2, strengthening application security while enabling seamless single sign-on (SSO) for employees.
- Refactored legacy APIs to reduce technical debt by rewriting inefficient code, improving performance, and following best practices. This enhanced system reliability and maintainability.
- Deployed applications on AWS using ECS, Lambda, and RDS, reducing infrastructure costs by 20% through optimized resource allocation.
- Built CI/CD pipelines with Jenkins and GitHub Actions, automating build, test, and deployment workflows. This reduced deployment time from 1 hour to 10 minutes.
- Implemented role-based access control (RBAC), ensuring restricted access to critical features based on user roles, improving security compliance.
- Containerized applications using Docker and managed deployments with Kubernetes, increasing deployment efficiency and reducing downtime.
- Configured Prometheus and Grafana dashboards for real-time monitoring, enabling proactive issue detection and reducing system outages.
- Developed automated background jobs using BullMQ and Redis, improving system efficiency by handling periodic tasks asynchronously.
- Enhanced frontend performance in Angular by implementing lazy loading and optimizing state management with NgRx, reducing initial load times by 35%.
- Collaborated with cross-functional teams in an Agile Scrum environment, participating in sprint planning, stand-ups, and retrospectives to ensure timely project delivery.

Nationwide Insurance: Jon Peters Insurance

Aug 2023 - May 2024

Full Stack Developer

Athens, Alabama, USA

- Developed and maintained RESTful APIs using Node.js and Express.js, enabling efficient data exchange between the policy management system and external services. This streamlined policy handling and reduced manual processing.
- Built a React-based claims management dashboard, allowing insurance agents to view and process claims in real time. This improved claim resolution speed by 30%.
- Implemented WebSockets for real-time claim status updates, notifying customers and agents instantly about claim progress, reducing support queries by 25%.

- Integrated Stripe and PayPal payment gateways, allowing customers to securely pay premiums online. This reduced payment failures and increased successful transactions.
- Developed a background job processing system using BullMQ and Redis, automating policy renewal reminders and reducing manual workload.
- Optimized PostgreSQL queries, reducing query execution time for policy reports from 3 seconds to under 500ms by adding indexes and refactoring queries.
- Implemented JWT authentication and session management, ensuring secure access to the policyholder portal.
- Automated document generation using Puppeteer and Node.js, eliminating manual report creation and reducing processing time.
- Enhanced security by setting up API Gateway with rate limiting and IP whitelisting, preventing DDoS attacks and unauthorized access.
- Built and deployed applications on AWS using Terraform, automating infrastructure provisioning and reducing deployment errors.
- Configured Elasticsearch for full-text search, allowing users to search claims and policies efficiently.
- Developed CI/CD pipelines using GitHub Actions, automating testing and deployment workflows for faster release cycles.
- Implemented GraphQL APIs, providing flexible and optimized data fetching capabilities for frontend applications.
- Used OpenTelemetry for distributed tracing, improving performance monitoring and debugging in microservices.
- Collaborated with stakeholders to understand business requirements, translating them into technical solutions and delivering feature enhancements on schedule.

Thomson Reuters

Feb 2020 - Aug 2022

Full Stack Developer

Hyderabad, India

- Developed scalable RESTful APIs in Node.js and Express.js, handling financial transaction data securely and efficiently.
- Built an Angular-based financial reporting system, enabling real-time financial insights for corporate clients.
- Implemented an event-driven architecture using Kafka, allowing seamless processing of large financial datasets asynchronously.
- Designed and integrated OAuth2-based authentication, ensuring secure access to financial APIs for third-party applications.
- Integrated third-party APIs for currency exchange rates, automating real-time conversion calculations for financial transactions.
- Optimized database performance for MySQL and PostgreSQL, improving query efficiency and reducing response times.
- Developed scheduled batch jobs using Node.js and AWS Lambda, automating periodic financial report generation.
- Refactored existing API endpoints, reducing response times and improving data consistency across services.
- Implemented API logging and monitoring with ELK Stack (Elasticsearch, Logstash, Kibana), improving error diagnosis and system observability.
- Integrated AWS S3 for secure document storage, ensuring financial reports remained accessible and backed up.
- Developed a notification system using WebSockets, alerting users of account updates and transaction approvals in real time.
- Improved frontend performance using Angular's lazy loading and preloading strategies, enhancing the user experience.
- Implemented JWT-based user authentication, strengthening security across financial applications and reducing unauthorized access incidents.
- Monitored system performance using New Relic and Prometheus, proactively identifying bottlenecks and optimizing backend services.
- Collaborated with DevOps teams to containerize applications with Docker and deploy them on Kubernetes clusters, ensuring high availability and scalability.

EDUCATION

Wright State University

Apr 2024

Master's, Computer Science